Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	CSC4220
Course title (English)	Matrix Computation
Course title (Chinese)	矩阵计算
Units	3
Language of Instruction	English
Description (English)	This is a hands-on course on numerical linear algebra and matrix computations emphasizing computational techniques dealing with algorithmic methodologies that are critical for applications in machine learning, optimization, statistics and scientific computing. Matrix computation methods complement algorithms focusing on discrete objects such as lists, trees and graphs, and those methods are becoming ever more critical with the rapid advances in AI technology. The topics covered include algorithms for solving linear systems, linear least squares problems, eigenvalue and singular value problems as well as iterative methods such as Arnoldi methods, Krylov subspace methods (including GMRES and preconditioned CG). Fundamental ideas of conditioning and numerical stability will also be covered in detail. In addition, several modern algorithmic paradigms such as randomization techniques will also be covered.
Description (Chinese)	这是一门关于数值线性代数和矩阵计算的实践课程，重点讲解与算法方法相关的计算技术，这些方法对机器学习、优化、统计和科学计算等领域的应用至关重要。矩阵计算方法补充了专注于离散对象（例如列表、树和图）的算法，随着人工智能技术的快速发展，这些方法正变得越来越重要。课程涵盖的主题包括用于求解线性系统、线性最小二乘问题、特征值和奇异值问题的算法，以及迭代方法，例如 Arnoldi 方法、Krylov 子空间方法（包括 GMRES 和预条件 CG）。课程还将详细介绍条件作用和数值稳定性的基本概念。此外，还将介绍一些现代算法范式，例如随机化技术。

2. Prerequisites / Co-requisites

A. Prerequisites

MAT 2040, MAT 2041, MAT 2041A

B. Co-requisites

3. Learning Outcomes

• Understanding of the concepts and methodology of numerical linear algebra.
• Proficiency in the use of high-level language for matrix computation algorithms, such as Matlab or Python.
• Knowledge of matrix decompositions, such as LU, QR, Cholesky, Schur and SVD, and the their computational algorithms
• Understanding of the theory of numerical linear algebra, such as perturbation analysis, backward error analysis, stability and convergence analysis.

4. Course syllabus

Linear algebra basics
1 Matrix-vector multiplication, orthogonality
2 Norms and SVD
Linear least squares problems
3 Projections and QR factorization
4 Householder triangulation and linear least squares methods
Conditioning and numerical stability
5 Condition numbers and
oating-point arithmetic
6 Numerical stability, conditioning of least squares problems
Linear systems
7 Gaussian elimination and pivoting
8 Cholesky factorization
Eigenvalue problems
9 Hessenberg reduction and QR algorithms
10 QR algorithms with shifts, and SVD computation
Iterative methods
11 Arnoldi iteration
12 GMRES
13 Lanczos methods

14 Preconditioning

5. Assessment Scheme

Component/ method	**% weight**
Assignments (including mini software projects)	20
Mid-Term Exam	35
Final Exam	45

6. Grade descriptor

General

Grade	**Description**
A	• Excellent understanding of the concepts of numerical linear algebra concepts. • Excellence in matrix computation algorithms. • Excellent knowledge of matrix decompositions, such as LU, QR, Cholesky, Schur, and SVD. • Excellent understanding of the major theory, such as such as perturbation analysis, backward error analysis, stability and convergence analysis. • Excellent in applying the knowledge and skills of numerical linear algebra in various science and engineering domains. • Excellent in using Matlab or Python for numerical linear algebra algorithms.
B	• Good understanding of the concepts of numerical linear algebra concepts. • Excellence in matrix computation algorithms. • Good knowledge of matrix decompositions, such as LU, QR, Cholesky, Schur, and SVD. • Good understanding of the major theory, such as such as perturbation analysis, backward error analysis, stability and convergence analysis. • Good in applying the knowledge and skills of numerical linear algebra in various science and engineering domains. • Good in using Matlab or Python for numerical linear algebra algorithms.
C	• Understanding of most of the concepts of numerical linear algebra concepts. • Familiarity in matrix computation algorithms. • Basic knowledge of matrix decompositions, such as LU, QR, Cholesky, Schur, and SVD. • Good in using Matlab or Python for numerical linear algebra algorithms.
D	• Basic understanding of most of the concepts of numerical linear algebra concepts.

	• Familiarity in matrix computation algorithms. • Some knowledge of matrix decompositions, such as LU, QR, Cholesky, Schur, and SVD.
F	• Limited understanding of the basic concepts of numerical linear algebra • Non-ability of doing basic matrix computations algorithms

7. Feedback for evaluation

• Communicate with instructor and/or TAs by chatting, Wechat, blackboard, emails etc.
• Evaluate students' performance through assignments, mid-term and final exams.
• Collect students' feedback from CTE by the end of the course.
• Collect opinions from peers by internal and external course reviews.

8. Reading

Required Reading

1.Numerical Linear Algebra, 0898713617, Trefethen, Bau, TBD, United States

9. Course components

Activity	**Hours/week**
Lectures	3
Tutorials	1

10. Indicative teaching plan

Week	**Content/ topic/ activity**
1	Matrix-vector multiplication, orthogonality
2	Norms and SVD
3	Projections and QR factorization
4	Householder triangulation and linear least squares methods
5	Condition numbers and oating-point arithmetic
6	Numerical stability, conditioning of least squares prob
7	Gaussian elimination and pivoting

8	Cholesky factorization
9	Hessenberg reduction and QR algorithms
10	QR algorithms with shifts, and SVD computation
11	Arnoldi iteration
12	GMRES
13	Lanczos methods
14	Preconditioning

11. Any other information

12. Version date

Version number	1
As of (date)	2025/06/12